Project Proposal

Project: Real-Time Chemotherapy Patient Monitoring with AI-Based Anomaly Detection

Objective: Build and evaluate a reproducible, in-hospital chemotherapy monitoring prototype that processes multi-source events and flags unusual vital-sign patterns using Isolation Forest. The output supports clinician review; it does not provide diagnoses or calculate doses.

Description: Simulated monitoring devices supply heart rate and temperature. Nursing staff are represented by synthetic pain/nausea entries; laboratory events provide ANC and selected CBC results. A single pipeline links events to patients and treatment sessions, handles missing/duplicate/late data, shows trends and alerts, and keeps clinician-entered review and dose-change history. Home monitoring is excluded; full appointment scheduling is deferred beyond the core deliverables.

Tools & Design: Python-generated synthetic observations -> Kafka -> Spark Structured Streaming validation/event-time processing -> PostgreSQL storage. A lightweight Python worker extracts five-minute heart-rate/temperature features and applies a pre-trained, saved Isolation Forest (scikit-learn); Streamlit displays the resulting flags and an independent rules-based alert stream. Initial training and calibration occur offline; the saved model is reused without live retraining or LSTM.

Technologies to Use: Python, Apache Kafka, Apache Spark Structured Streaming, PostgreSQL, Streamlit, scikit-learn (IsolationForest), Pandas/NumPy, Git; local development tools and optional containers.

Team Members & Responsibilities (individual allocation - backup option):

1- Member 1 - Name: **Nasser Shaaban** | Data Simulation & Kafka: Synthetic patient/session scenarios, labels and split manifests; Kafka producers/topics, replay fixtures and delivery tests.

2- Member 2 - Name: **Kholod Ismael** | Spark Streaming & Data Quality: Spark ingestion, schema/unit/time checks, duplicate and late-event handling; persistence integration, counters and recovery tests.

3- Member 3 - Name: **Ziad Ashraf** | Database & Feature Engineering: PostgreSQL schema/access helpers and integrity; shared five-minute feature calculations, eligibility and offline/online parity tests.

4- Member 4 - Name: **Sohair Asaad** | Anomaly Detection & Evaluation: Isolation Forest training/calibration, saved-model scoring worker and versions; held-out evaluation and AI failure tests.

5- Member 5 - Name: **Samaa Hassan** | Dashboard & Review Interface: Streamlit patient/session views, trends, laboratory context and AI/data status; review forms, history display and UI tests.

6- Member 6 - Name: **Ayah Mossad** | Rule Alerts & Review Records: Independent rules, evidence-linked alerts/status and review-record service; append-only decisions and workflow/backend tests.

Stage 1: First Submission — Due 16 October 2026

- **Nasser Shaaban: prepare planning objectives, scope, source-data assumptions and stakeholder needs.**
- **Kholod Ismael: prepare planning stages, responsibilities, technical dependencies and risk log.**
- **Ziad Ashraf: review eSMART and Zablah; draft their summaries, limits and engineering lessons.**
- **Sohair Asaad: review Mallesh; complete literature comparison, AI rationale and references.**
- **Samaa Hassan: draft requirements steps 1-4, functional needs, data fields and user workflows.**
- **Ayah Mossad: draft requirements steps 5-7, priorities and acceptance checks; all members cross-review.**

Deliverables:

- Project Proposal / Project Planning & Management, following the supplied proposal headings.
- Literature Review: comparison, research gap, methodology rationale and references.
- Requirements Gathering: stakeholders, requirements, priorities and verification plan.
- Official deadline: submit all three documents on 16 October 2026.

Stage 2: System Analysis & Design — Due 6 November 2026

- **Nasser Shaaban: design synthetic data, patient splits, Kafka topics, message contracts and replay fixtures.**
- **Kholod Ismael: design Spark validation, event-time handling, duplicate control and recovery checks.**
- **Ziad Ashraf: design PostgreSQL tables/access, shared feature calculations and window eligibility.**
- **Sohair Asaad: design Isolation Forest calibration, saved-model scoring and held-out evaluation.**
- **Samaa Hassan: design dashboard views, patient/session filters, review forms and status displays.**
- **Ayah Mossad: design independent rules, alert evidence/status and append-only review-record service.**

Deliverables:

- Final architecture, data-flow diagram and Kafka–Spark interface specification.
- Logical database model, data dictionary and synthetic-data generation specification.
- Isolation Forest design and evaluation protocol; dashboard wireframes.
- Official deadline: submit System Analysis & Design on 6 November 2026.

Stage 3: Implementation & Execution — Due 30 November 2026

- **Nasser Shaaban: implement simulator/Kafka publishing supply reproducible labels, splits and replay tests.**
- **Kholod Ismael: implement Spark validation, late/duplicate handling and persistence; test recovery.**
- **Ziad Ashraf: implement database/helpers and shared feature code; test integrity and feature parity.**
- **Sohair Asaad: train/calibrate the model, implement scoring and evaluate untouched test scenarios.**
- **Samaa Hassan: implement dashboard, review forms and history views; test refresh and visibility.**
- **Ayah Mossad: implement rule alerts and review-record service; test history. All members integrate/test.**

Deliverables:

- Working source code for the data generator, Kafka/Spark pipeline and database.
- Saved Isolation Forest model, evaluation results, rule alerts and dashboard.
- Test evidence for abnormal, missing, duplicate and late events; setup instructions.
- Official deadline: submit Implementation (Source Code & Execution) on 30 November 2026.

Stage 4: Final Presentation, Testing & Reports — Due 4 December 2026

- **Nasser Shaaban: demonstrate synthetic scenarios, Kafka publication and repeatable delivery fixtures.**
- **Kholod Ismael: present streaming validation, message reconciliation and recovery findings.**
- **Ziad Ashraf: present database integrity, evidence links and verified feature-window calculations.**
- **Sohair Asaad: present model evaluation, failure behaviour and synthetic-data limitations.**
- **Samaa Hassan: demonstrate dashboard, review forms and history; explain interface test results.**
- **Ayah Mossad: demonstrate rules and decision records; all six assemble evidence and rehearse.**

Deliverables:

- Final project presentation and live or recorded system demonstration.
- Testing report: functional checks, pipeline latency, reliability and AI results.
- Final written report, source-code references and reproducibility instructions.
- Official deadline: submit the final presentation, testing and reports on 4 December 2026.

Official DEPI submission deadlines (four milestones):

Deadline	Required submission
16 October 2026	Project Planning & Management + Literature Review + Requirements Gathering
6 November 2026	System Analysis & Design
30 November 2026	Implementation (Source Code & Execution)
4 December 2026	Final Presentation, Testing & Reports